TASK – 10

Secure Authentication Stress Testing Using Java

PROGRAM :

```
import java.util.concurrent.*;
import java.util.concurrent.atomic.AtomicInteger;

public class AuthenticationStressTest {

   static final String USER = "admin";
   static final String PASS = "Secure123";

   static boolean authenticate(String username, String password) {
      return USER.equals(username) && PASS.equals(password);
   }

   public static void main(String[] args)
         throws InterruptedException {

      int totalRequests = 1000;
      int threads = 10;

      ExecutorService executor =
            Executors.newFixedThreadPool(threads);

      AtomicInteger success = new AtomicInteger();
      AtomicInteger failure = new AtomicInteger();
```

```
long start = System.currentTimeMillis();

for (int i = 0; i < totalRequests; i++) {
    final int request = i;

    executor.submit(() -> {
        boolean valid;

        if (request % 2 == 0) {
            valid = authenticate("admin", "Secure123");
        } else {
            valid = authenticate("admin", "WrongPass");
        }

        if (valid) {
            success.incrementAndGet();
        } else {
            failure.incrementAndGet();
        }
    });
}

executor.shutdown();
executor.awaitTermination(1, TimeUnit.MINUTES);

long end = System.currentTimeMillis();

System.out.println("--- Stress Test Results ---");
```

```
        System.out.println("Total Requests: " + totalRequests);
        System.out.println("Successful Logins: " + success.get());
        System.out.println("Failed Logins: " + failure.get());
        System.out.println("Execution Time: "
                + (end - start) + " ms");

    }
}
```

INPUT :

Total Requests: 1000

Threads: 10

Valid login attempts: 500

Invalid login attempts: 500

OUTPUT :

--- Stress Test Results ---

Total Requests: 1000

Successful Logins: 500

Failed Logins: 500

Execution Time: 25 ms